



Mastering Observability with OS Tools



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Who Am I

- **Hi** 🖐️ I'm Luca Raveri, Software Engineer and AWS Solutions Architect



💻 Backend Development

☁️ Cloud Architectures

🔍 Observability

📌 Venice, Italy



**Do you
really know
your system?**



Common Scenario

The day after the release...



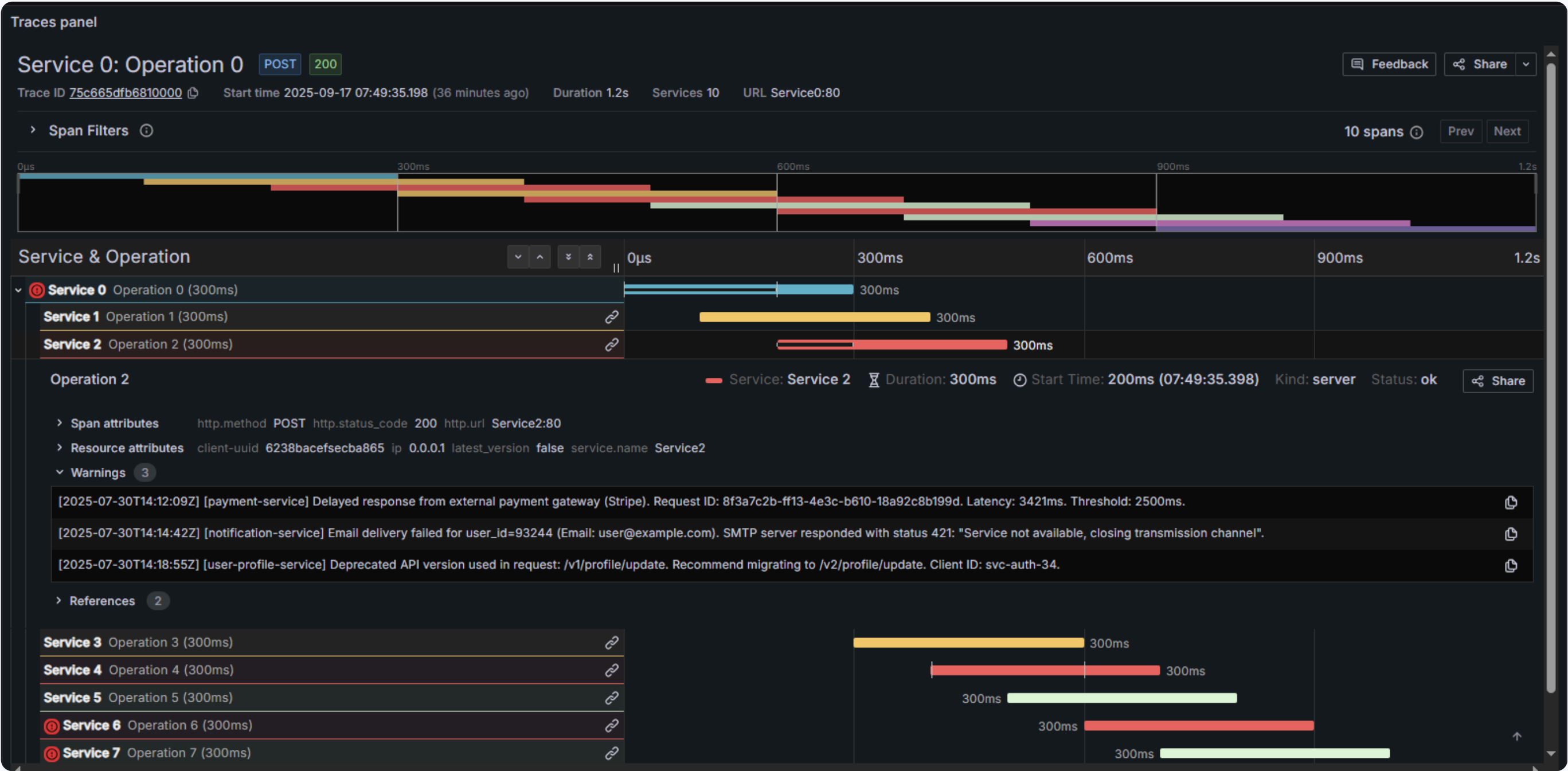
What are Monitoring and Observability?

- **Monitoring**: Continuous measurement of a system's health to detect anomalies, performance issues, or downtime → *tells us that something is wrong*
- **Observability**: The ability to understand the internal state of a complex system by examining the data it produces → *tells us why it is wrong*

The Three Pillars Of Observability

- **Metrics:** Numeric measurements over time (e.g. CPU usage)
→ *What is happening*
- **Logs:** Discrete, timestamped records of events → *Why is happening*
- **Traces:** Request flow through the system → *Where is happening*

What is a Trace



Key Benefits of Observability

- **Lower MTTD (Mean Time to Detect)**

Problems are identified quickly, often before users notice.

- **Lower MTTR (Mean Time to Resolve)**

Root causes are found and fixed faster, reducing downtime and impact

From This...

```
[2025-09-30 19:05:37.130 +0200] INFO: service boot {"service":"grafana-demo","payload":{"requestId":"da34f8bf-0bfa-435e-8198-cfdf016b9de8","service":"orders-api","e
[2025-09-30 19:05:37.370 +0200] INFO: incoming request {"service":"grafana-demo","payload":{"requestId":"6d3a0a0c-0edb-4e7d-badf-0aa99447f19f","method":"GET","path"
[2025-09-30 19:05:37.418 +0200] INFO: request completed {"service":"grafana-demo","payload":{"requestId":"6d3a0a0c-0edb-4e7d-badf-0aa99447f19f","method":"GET","path"
[2025-09-30 19:05:37.529 +0200] DEBUG: sql execution {"service":"grafana-demo","payload":{"requestId":"2da464f1-2bf9-455b-8efa-60ee5ba10585","sql":"SELECT 1","db":"
[2025-09-30 19:05:37.576 +0200] INFO: incoming request {"service":"grafana-demo","payload":{"requestId":"152e854c-64a5-43e4-b8fe-f3b45cf97a4a","method":"POST","path"
[2025-09-30 19:05:37.696 +0200] DEBUG: sql execution {"service":"grafana-demo","payload":{"requestId":"152e854c-64a5-43e4-b8fe-f3b45cf97a4a","sql":"insert into user
[2025-09-30 19:05:37.739 +0200] INFO: request completed {"service":"grafana-demo","payload":{"requestId":"152e854c-64a5-43e4-b8fe-f3b45cf97a4a","method":"POST","pat
[2025-09-30 19:05:37.988 +0200] ERROR: rate limit nearing {"service":"grafana-demo","payload":{"requestId":"4be049ab-ca57-417e-94d1-147fab6b1fd7","clientId":"demo-c
[2025-09-30 19:05:38.049 +0200] INFO: incoming request {"service":"grafana-demo","payload":{"requestId":"ea90cf08-7d69-4e75-89f6-af7e87e96ef2","method":"GET","path"
[2025-09-30 19:05:38.124 +0200] DEBUG: sql execution {"service":"grafana-demo","payload":{"requestId":"ea90cf08-7d69-4e75-89f6-af7e87e96ef2","sql":"select id, email
[2025-09-30 19:05:38.163 +0200] INFO: request completed {"service":"grafana-demo","payload":{"requestId":"ea90cf08-7d69-4e75-89f6-af7e87e96ef2","method":"GET","path"
[2025-09-30 19:05:38.206 +0200] INFO: incoming request {"service":"grafana-demo","payload":{"requestId":"fb0b4ea8-1050-45a7-ac61-9bad34bc265f","method":"PATCH","pat
[2025-09-30 19:05:38.281 +0200] DEBUG: sql execution {"service":"grafana-demo","payload":{"requestId":"fb0b4ea8-1050-45a7-ac61-9bad34bc265f","sql":"update users set
[2025-09-30 19:05:38.316 +0200] INFO: request completed {"service":"grafana-demo","payload":{"requestId":"fb0b4ea8-1050-45a7-ac61-9bad34bc265f","method":"PATCH","pa
[2025-09-30 19:05:38.455 +0200] ERROR: unhandled application error {"service":"grafana-demo","payload":{"requestId":"d4d9fafa-b284-416f-baba-2a1dfbb6f85b","method":
[2025-09-30 19:05:38.562 +0200] DEBUG: sql execution {"service":"grafana-demo","payload":{"requestId":"d4d9fafa-b284-416f-baba-2a1dfbb6f85b","sql":"select id from u
[2025-09-30 19:05:38.600 +0200] INFO: request completed {"service":"grafana-demo","payload":{"requestId":"d4d9fafa-b284-416f-baba-2a1dfbb6f85b","method":"GET","path"
[2025-09-30 19:05:38.692 +0200] INFO: incoming request {"service":"grafana-demo","payload":{"requestId":"cbca0ba1-cae2-4555-a516-62bb5329a166","method":"GET","path"
[2025-09-30 19:05:38.756 +0200] DEBUG: cache get {"service":"grafana-demo","payload":{"requestId":"cbca0ba1-cae2-4555-a516-62bb5329a166","key":"order:5012","backend
[2025-09-30 19:05:38.781 +0200] INFO: request completed {"service":"grafana-demo","payload":{"requestId":"cbca0ba1-cae2-4555-a516-62bb5329a166","method":"GET","path"
[2025-09-30 19:05:38.905 +0200] ERROR: slow query detected {"service":"grafana-demo","payload":{"requestId":"25012bdd-1a6a-45e2-9d32-6f1b6c803e32","sql":"select * f
[2025-09-30 19:05:39.090 +0200] INFO: incoming request {"service":"grafana-demo","payload":{"requestId":"e1fef4d3-beec-4480-a0ca-9fb6782213fe","method":"DELETE","pa
[2025-09-30 19:05:39.131 +0200] INFO: request completed {"service":"grafana-demo","payload":{"requestId":"e1fef4d3-beec-4480-a0ca-9fb6782213fe","method":"DELETE","p
[2025-09-30 19:05:39.241 +0200] ERROR: payment gateway timeout {"service":"grafana-demo","payload":{"requestId":"12ceeb00-0a86-4ea0-bce0-99efade0a625","method":"POS
[2025-09-30 19:05:39.413 +0200] DEBUG: sql execution {"service":"grafana-demo","payload":{"requestId":"12ceeb00-0a86-4ea0-bce0-99efade0a625","sql":"insert into paym
```


...to This!

environmentproduction ▾

levelAll ▾

messageEnter value

dataEnter value

<<🕒2025-10-03 23:00:00 to 2025-10-04 08:59:59 ▾>>🔍

🔄 Refresh ▾

requestIdEnter value

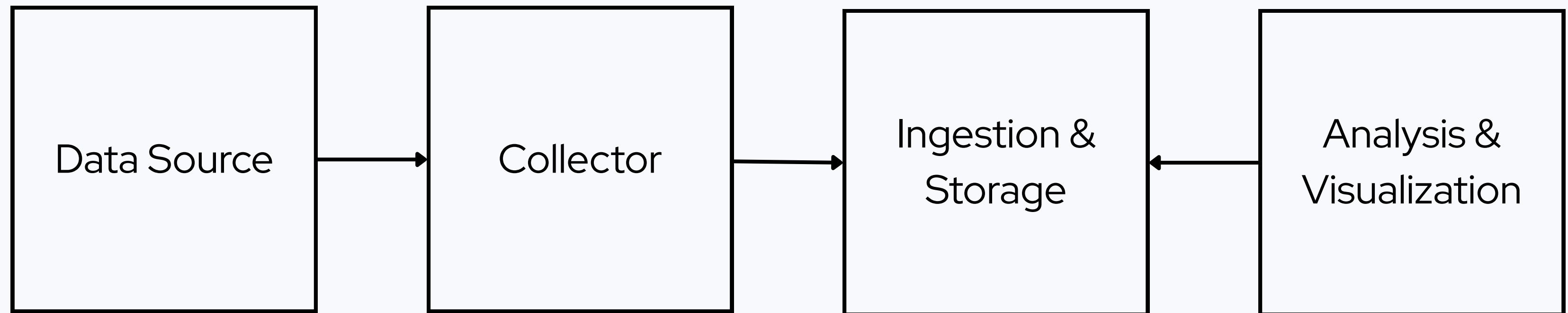
New panel

time ▾	level	requestId	msg	payload
2025-10-04 08:59:52	error	115a07458215	HTTP request failed	{"requestId":"115a07458215","method":"GET","endpoint":"/api/users/123","statusCode":400,"error":
2025-10-04 08:59:52	info	115a07458215	HTTP response sent	{"requestId":"115a07458215","method":"GET","endpoint":"/api/users/123","statusCode":400,"request
2025-10-04 08:59:52	debug	115a07458215	Database query executed	{"requestId":"115a07458215","operation":"SELECT","table":"users","query":"SELECT * FROM users W
2025-10-04 08:59:52	debug	115a07458215	Database query executed	{"requestId":"115a07458215","operation":"SELECT","table":"payments","query":"SELECT * FROM pay
2025-10-04 08:59:52	info	115a07458215	HTTP request received	{"requestId":"115a07458215","method":"GET","endpoint":"/api/users/123","userAgent":"Mozilla/5.0 (c
2025-10-04 08:59:51	error	ecf36bd28b6e	HTTP request failed	{"requestId":"ecf36bd28b6e","method":"PUT","endpoint":"/api/users/123","statusCode":404,"error":
2025-10-04 08:59:51	info	ecf36bd28b6e	HTTP response sent	{"requestId":"ecf36bd28b6e","method":"PUT","endpoint":"/api/users/123","statusCode":404,"reques
2025-10-04 08:59:51	debug	ecf36bd28b6e	Database query executed	{"requestId":"ecf36bd28b6e","operation":"SELECT","table":"orders","query":"SELECT * FROM order:
2025-10-04 08:59:51	debug	ecf36bd28b6e	Database query executed	{"requestId":"ecf36bd28b6e","operation":"SELECT","table":"payments","query":"SELECT * FROM pa
2025-10-04 08:59:50	info	ecf36bd28b6e	HTTP request received	{"requestId":"ecf36bd28b6e","method":"PUT","endpoint":"/api/users/123","userAgent":"Mozilla/5.0 (
2025-10-04 08:59:49	info	a944209e12b9	HTTP response sent	{"requestId":"a944209e12b9","method":"DELETE","endpoint":"/api/users/123","statusCode":200,"rec
2025-10-04 08:59:49	debug	a944209e12b9	Database query executed	{"requestId":"a944209e12b9","operation":"SELECT","table":"payments","query":"SELECT * FROM pa
2025-10-04 08:59:49	debug	a944209e12b9	Database query executed	{"requestId":"a944209e12b9","operation":"UPDATE","table":"orders","query":"UPDATE * FROM orde
2025-10-04 08:59:49	info	a944209e12b9	HTTP request received	{"requestId":"a944209e12b9","method":"DELETE","endpoint":"/api/users/123","userAgent":"Mozilla/

Demo Time!



Let's Build an Observability System



LGTM Stack

Loki



Database
for logs

Grafana



Visualization
tool

Tempo



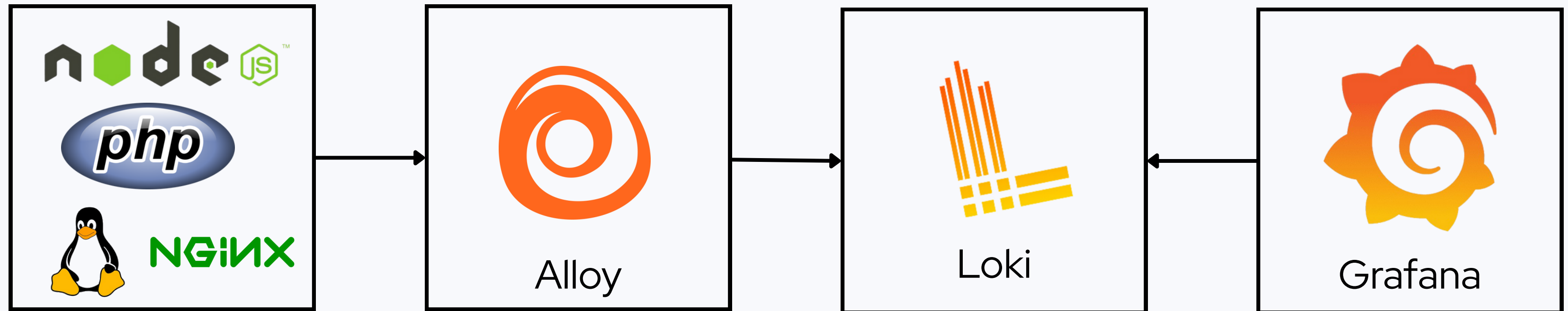
Database
for traces

Mimir



Database
for metrics

Let's build an Observability System



Let's Start Logging

- Use a logger instead of native functions
- Apply log levels and timestamps
- Format logs (JSON)
- Never log secrets and PII
- Set retention policies (e.g., logrotate)

Logging Strategies

- Log system inputs and outputs
- Centralize logging in a single layer
- Use UUIDs to correlate related logs
- More logs means more insight, but higher overhead



```
console.log(`Customer ${customerName} purchased ${itemsPurchased} items`);
```



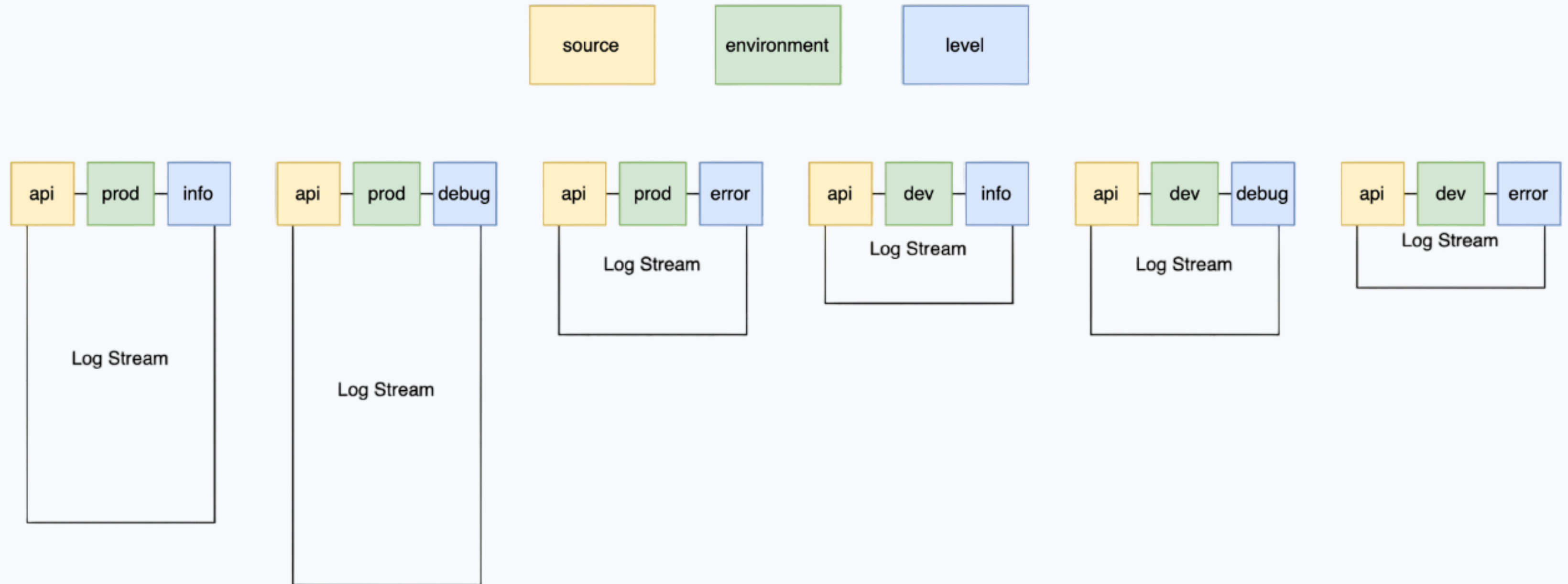
```
logger.info('Customer purchase', {  
  customerId: customer.id,  
  customerName: customer.name,  
  itemsPurchased: customer.itemsPurchased,  
  totalAmount: customer.totalAmount,  
});
```



What is Loki

“Loki is a horizontally scalable, highly available, multi-tenant log aggregation system inspired by Prometheus. It is designed to be very cost effective and easy to operate. It does not index the contents of the logs, but rather a set of labels for each log stream.”

How Loki Works



"...It does not index the contents of the logs, but rather a set of labels for each log stream."

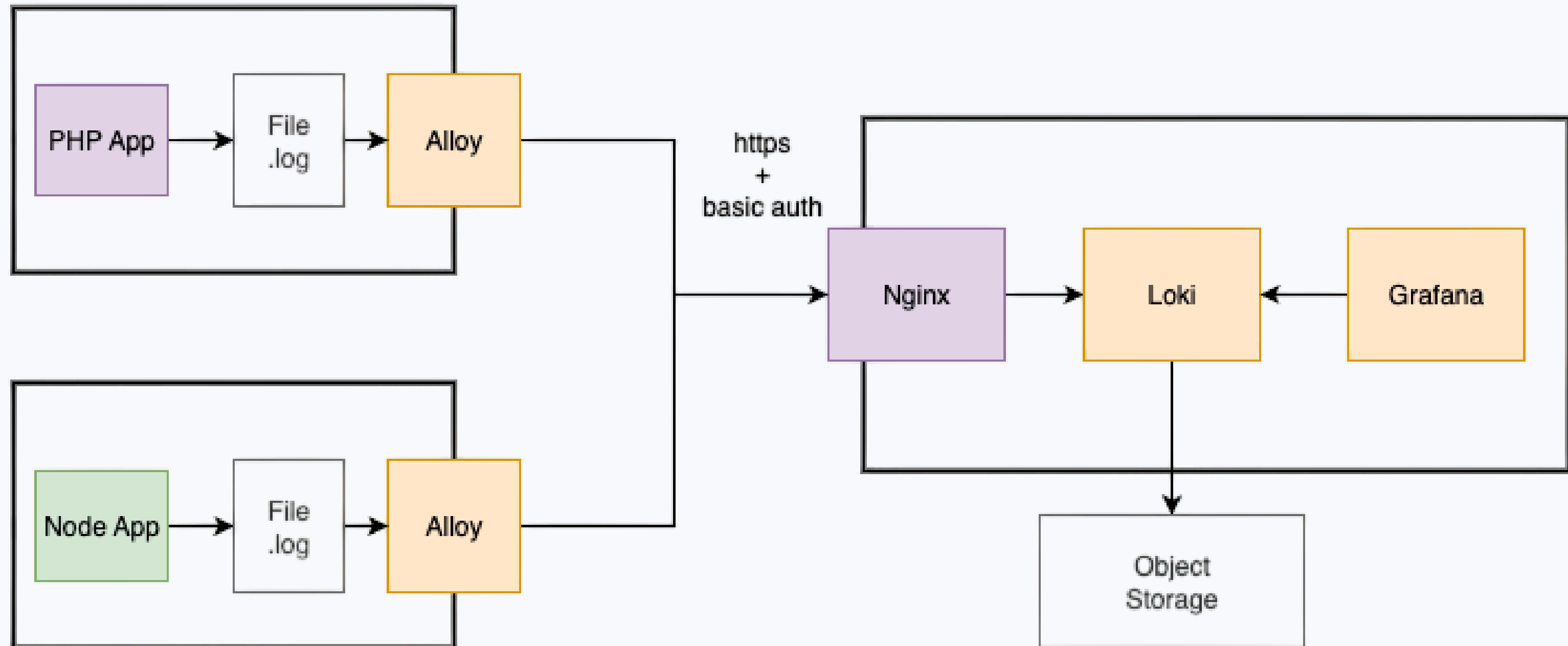
How to Send Logs to Loki

Grafana Alloy is a collector that automatically reads your log files and pushes them to Loki. It can:

- Apply **static labels** to all logs from a file
- Extract **dynamic labels** from log content
- Perform **transformations** on log data



Hosting Loki



How to Query Logs

```
{source="grafana-demo", environment=~"${environment}", level=~"${level}"
|~ "${data}"
| json
| msg =~ ".*${message}.*"
| payload_requestId =~ ".*${requestId}.*"
```

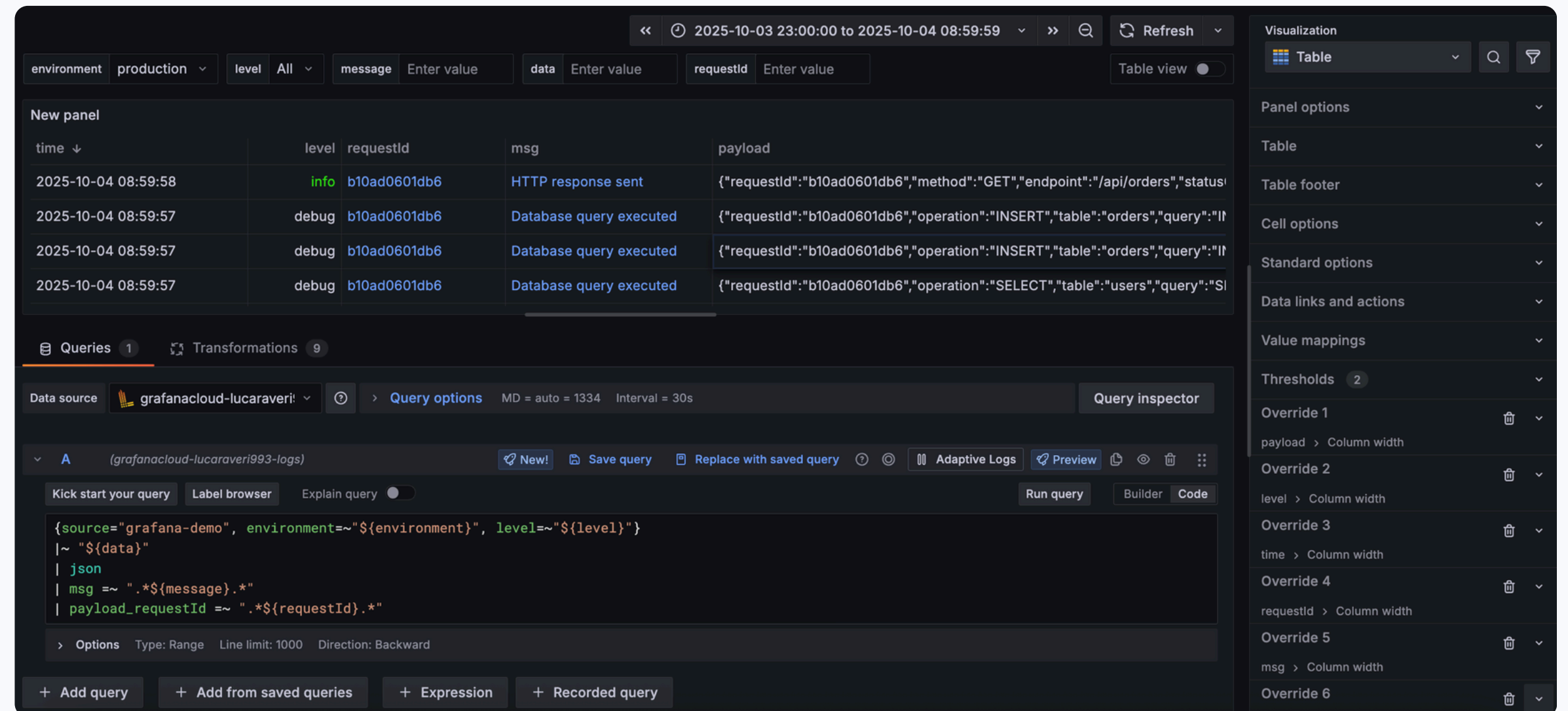


```
{job="apache_error_log", environment=~"${environment}", source=~"${source}"}  
|~ "${data}"  
| regexp "\\[(?P<timestamp>[^\]]+)]\\.\\.[php7:(?P<level>[^\]]+)\\.\\.\\.PHP (?P<type>[:]+): (?P<message>.*) in (?P<file>[A-Z]:\\\\\\\\[^\"']+)(?:.(?P<line>\\\\d+)| on line (?P<line_alt>\\\\d+))"  
| line_format "{ printf `{"timestamp": \"%s\", \"level\": \"%s\", \"type\": \"%s\", \"message\": \"%s\",  
\"file\": \"%s\", \"line\": \"%s\"}` .timestamp .level .type .message .file (or .line .line_alt) } }"  
| message =~ ".$${message}."  
| file =~ ".$${file}."  
| message != ""
```



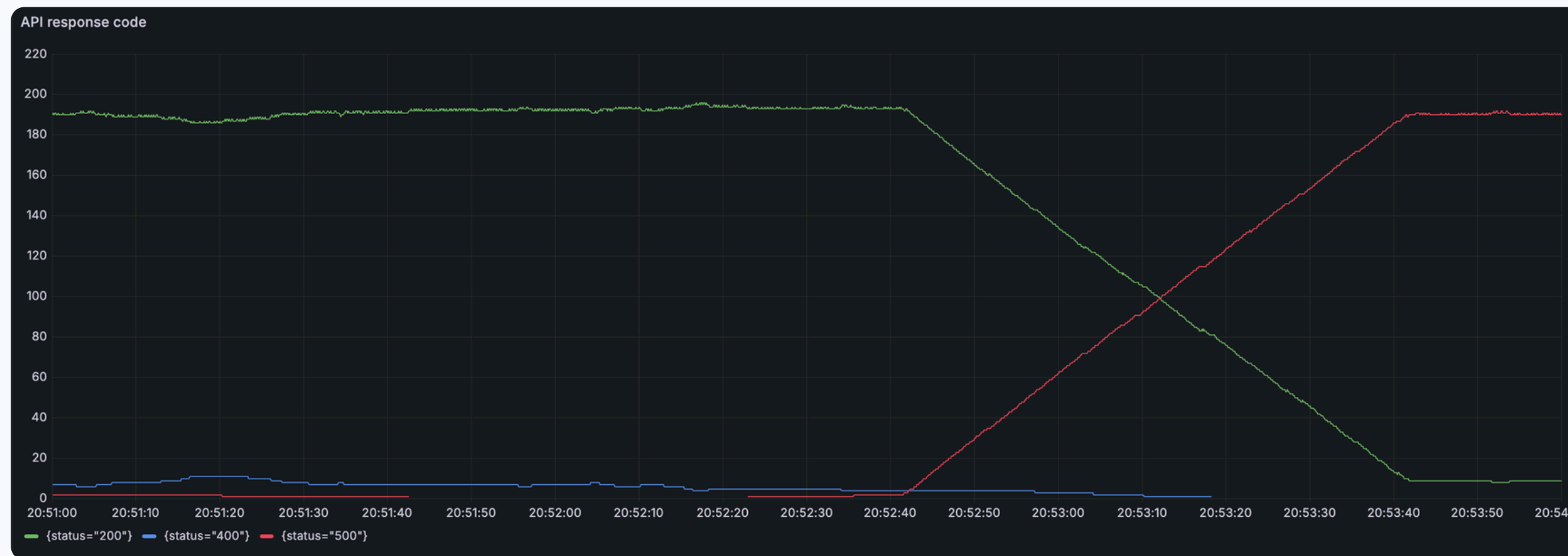
How to Visualize Logs

1. Write the query
2. Apply transformations
3. Create variables
4. Add overrides



Alerting Strategies

- Single error detection → trigger on critical errors
- Error rate spikes (RED method) → alert on sudden increase in errors



Results

- Strong team adoption and enthusiasm.
- Significantly improved **MTTD** and reduced **MTTR** from 2 days to 2 hours, enhancing service quality.
- Handled 1 GB/day with 2-week retention (14 GB total), running smoothly on a €6/month instance.

Thank you!

Questions?

Demo Repository

